Problem 4

Part (a)

Let $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mathbb{P}_\lambda)$ be the probability space, where $\mathbb{P}_\lambda$ is the probability measure defined by the density function $f_\lambda(y) = \lambda e^{-\lambda y} 1_{[0,\infty)}(y)$.

Using the distribution function $F_\lambda(y) = \mathbb{P}_\lambda((-\infty, y]) = 1 - e^{-\lambda y}$ for $y \geq 0$ and the complement rule, we determine the probability of the event (y, ∞) for any $y \geq 0$:

$$\mathbb{P}_\lambda((y,\infty)) = 1 - \mathbb{P}_\lambda((-\infty, y]) = 1 - (1 - e^{-\lambda y}) = e^{-\lambda y} \tag{2}$$

By the definition of conditional probability, for all $x, t > 0$, we have:

$$\mathbb{P}_\lambda((x+t,\infty) \mid (t,\infty)) = \frac{\mathbb{P}_\lambda((x+t,\infty) \cap (t,\infty))}{\mathbb{P}_\lambda((t,\infty))}$$

Because $x > 0$ and $t > 0$, it follows that $x + t > t$. Therefore, $(x+t,\infty) \subseteq (t,\infty)$, implying $(x+t,\infty) \cap (t,\infty) = (x+t,\infty)$. Then, for all $t, x > 0$, we have:

$$\begin{aligned}
\mathbb{P}_\lambda((x+t,\infty) \mid (t,\infty)) &= \frac{\mathbb{P}_\lambda((x+t,\infty))}{\mathbb{P}_\lambda((t,\infty))} \\
&\overset{(2)}{=} \frac{e^{-\lambda(x+t)}}{e^{-\lambda t}} = \frac{e^{-\lambda x} \cdot e^{-\lambda t}}{e^{-\lambda t}} = e^{-\lambda x} \\
&\overset{(2)}{=} \mathbb{P}_\lambda((x,\infty))
\end{aligned} \tag{3}$$

$\square$

Part (b) i)

Let $(\mathbb{R}_+, \mathcal{B}(\mathbb{R}_+), \mathbb{P})$ be the probability space, where $\mathbb{P}$ is an absolutely continuous probability measure. We define $G(t) := \mathbb{P}((t,\infty))$ for $t \geq 0$. By the memoryless property (1) and identical equivalences of (3) in Part (a), for all $t, x > 0$, we have:

$$\mathbb{P}((x,\infty)) = \frac{\mathbb{P}((x+t,\infty))}{\mathbb{P}((t,\infty))}$$

Substituting $G(\cdot)$ into this equation, we obtain for all $x, t > 0$:

$$G(x) = \frac{G(x+t)}{G(t)} \implies G(x+t) = G(x)G(t) \tag{4}$$

Additionally, since $\mathbb{P}$ is absolutely continuous on $\mathbb{R}_+$, $\mathbb{P}(\{0\}) = 0$. Thus, $G(0) = \mathbb{P}((0,\infty)) = 1 = G(1)^0$.

Step 1: Show $G(t) = G(1)^t$ for $t \in \mathbb{N}$. We proceed by induction over $n \in \mathbb{N}$.
Base Case ($n = 1$): $G(1) = G(1)^1$ holds trivially.
Induction Hypothesis (I.H.): Assume $G(n) = G(1)^n$ holds for a fixed $n \in \mathbb{N}$.
Induction Step ($n \to n+1$): Using equation (4) with $x = n$ and $t = 1$:

$$G(n+1) = G(n)G(1) \overset{\text{I.H.}}{=} G(1)^n \cdot G(1) = G(1)^{n+1}$$

Therefore, $G(n) = G(1)^n$ for all $n \in \mathbb{N}_{\nvdash}$.

Step 2: Show $G(t) = G(1)^t$ for $t \in \mathbb{Q}_+$. Let $t \in \mathbb{Q}_+$ such that $t = \frac{p}{q}$ for $p, q \in \mathbb{N}$. We evaluate $G(p)$ in two distinct ways.

First, since $p \in \mathbb{N}$, we apply the result from Step 1:

$$G(p) = G(1)^p \tag{5}$$

Second, we can rewrite $p = q \cdot \frac{p}{q}$ and apply equation (4) iteratively $q-1$ times:

$$\begin{aligned}
G(p) &= G\left(\underbrace{\frac{p}{q} + \cdots + \frac{p}{q}}_{q \text{ times}}\right) = G\left(\underbrace{\left(\frac{p}{q} + \cdots + \frac{p}{q}\right)}_{q-1 \text{ times}} + \frac{p}{q}\right) \\
&\overset{(4)}{=} G\left(\underbrace{\left(\frac{p}{q} + \cdots + \frac{p}{q}\right)}_{q-2 \text{ times}} + \frac{p}{q}\right) \cdot \left(G\left(\frac{p}{q}\right)\right)^1 \overset{(4)}{=} G\left(\underbrace{\left(\frac{p}{q} + \cdots + \frac{p}{q}\right)}_{q-3 \text{ times}} + \frac{p}{q}\right) \cdot \left(G\left(\frac{p}{q}\right)\right)^2 \\
&\overset{(4)}{=} \ldots \overset{(4)}{=} G\left(\frac{p}{q}\right) \cdot \left(G\left(\frac{p}{q}\right)\right)^{q-1} = \left(G\left(\frac{p}{q}\right)\right)^q
\end{aligned} \tag{6}$$

Equating (5) and (6) for $G(p)$ yields:

$$\left(G\left(\frac{p}{q}\right)\right)^q = G(1)^p \implies G\left(\frac{p}{q}\right) = (G(1)^p)^{\frac{1}{q}} = G(1)^{\frac{p}{q}}$$

Therefore, $G(t) = G(1)^t$ for all $t \in \mathbb{Q}_+$.

Step 3: Show $G(t) = G(1)^t$ for $t \in \mathbb{R}_+$. Since $\mathbb{P}$ is an absolutely continuous distribution, it has a probability density function. Therefore, its distribution function $F(t)$ is continuous (Lemma 1.5.3 b). Since $G(t) = 1 - F(t)$, G is also a continuous function on $\mathbb{R}_+$.

For any $t \in \mathbb{R}_+$, there exists a sequence of rational numbers $(q_n)_{n\in\mathbb{N}} \subset \mathbb{Q}_+$ such that $\lim_{n\to\infty} q_n = t$. By the continuity of G and the continuity of the exponential function, we conclude:

$$G(t) = G\left(\lim_{n\to\infty} q_n\right) = \lim_{n\to\infty} G(q_n) \overset{\text{Step 2}}{=} \lim_{n\to\infty} G(1)^{q_n} = G(1)^t \tag{7}$$

□

Part (b) ii)

Let $F(t)$ be the distribution function of $\mathbb{P}$. By Definition 1.4.1, $F(t) = \mathbb{P}((-\infty, t])$. Since the support of the probability measure $\mathbb{P}$ is $\mathbb{R}_+$, we have $\mathbb{P}((-\infty, 0)) = 0$. Thus, for all $t \geq 0$:

$$F(t) = \mathbb{P}([0, t]) = 1 - \mathbb{P}((t, \infty)) = 1 - G(t).$$

Using (7) in part b (i), we substitute $G(t)$:

$$F(t) = 1 - G(1)^t \quad \text{for } t \geq 0. \tag{8}$$

Because $G(1) = \mathbb{P}((1, \infty))$ is a probability, it must hold that $0 \leq G(1) \leq 1$. We analyze the boundary cases:

- **Assume $G(1) = 1$:** Then $F(t) = 1 - 1^t = 0$ for all $t \geq 0$. This implies $\lim_{t\to\infty} F(t) = 0$, which violates the asymptotic property of distribution functions ($\lim_{t\to\infty} F(t) = 1$). Thus, $G(1) \neq 1$.

- **Assume** $G(1) = 0$**:** Then $G(t) = 0^t = 0$ for all $t > 0$, implying $F(t) = 1$ for all $t > 0$. However, since $\mathbb{P}$ is an absolutely continuous probability measure, the probability of any singleton is zero, yielding $F(0) = \mathbb{P}(\{0\}) = 0$. This introduces a jump discontinuity at $t = 0$, contradicting the theorem that the distribution function of an absolutely continuous measure must be continuous everywhere. Thus, $G(1) \neq 0$.

Therefore, $0 < G(1) < 1$. Because $G(1)$ lies strictly between 0 and 1, we can uniquely define a constant $\lambda > 0$ such that:

$$\lambda := -\ln(G(1)) \implies G(1) = e^{-\lambda}.$$

Substituting $G(1) = e^{-\lambda}$ into (8), we obtain the distribution function:

$$F(t) = \begin{cases} 1 - e^{-\lambda t} & \text{for } t \geq 0 \\ 0 & \text{for } t < 0 \end{cases}$$

Because $\mathbb{P}$ is absolutely continuous, it admits a density function $f(t)$. We find this density by differentiating $F(t)$ where it is differentiable ($t \neq 0$):

$$f(t) = F'(t) = \begin{cases} \lambda e^{-\lambda t} & \text{for } t > 0 \\ 0 & \text{for } t < 0 \end{cases}$$

Using the indicator function, this can be written compactly as $f(t) = \lambda e^{-\lambda t} 1_{[0,\infty)}(t)$ and exactly matches the defined density of the exponential distribution. Therefore, $\mathbb{P}$ is an exponential distribution with parameter $\lambda = -\ln(G(1)) = -\ln(\mathbb{P}((1,\infty)))$.

$\square$